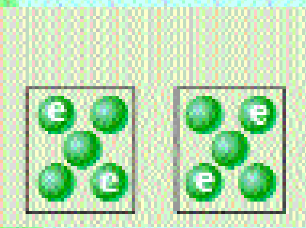
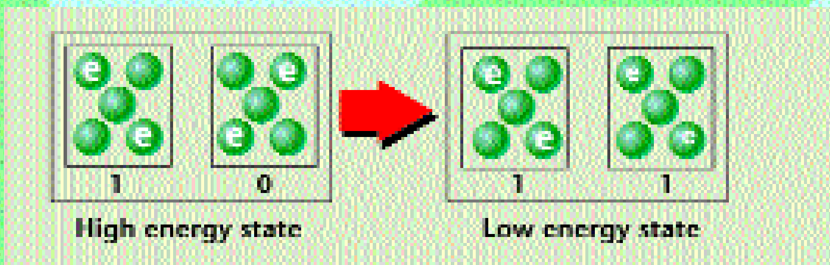
Stringed Quantum Dot Cells

An more interesting approach towards shrinkage is using quantum dot cells (QDCs): A quantum dot (QD), is like a tiny box that contains a certain number of electrons, and a QDC is a box that contains, say, five quantum dots, as below. Electrons repel each other, and the QDs that contain electrons move to opposite corners of the cell. This means they can be in one of two states, as below. When two QDCs of different states are placed next to each other, they are in a high energy state, because of their polarization properties. They tend to get into the low energy state by becoming of the same 'type', that is, with both of them having their electron-containing QDs at the same corners. The high and low energy states of pairs of QDCs can be 1s and 0s.

Using this property, a circuit without any 'real' current can be made by stringing QDCs together: polarize one in a particular way, and the others, like dominoes, become polarized the same way.



It turns out that logic gates, such as AND and OR, can be built using stringed QDCs, without the need for transistors. These gates can be very small: A single logic gate made of transistors, as they would be in 2010, would take up forty times the space than one made of QDCs would.

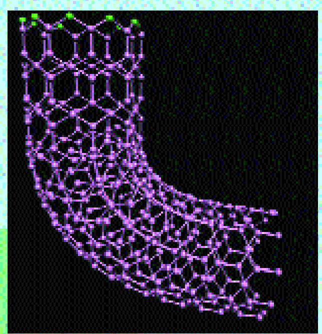


Nanotubes and NRAM

Applications of QDCs are still some time away. Closer to today, Nano RAM (NRAM), being developed by a company called Nantero, based in Woburn, Massachusetts, may be out any time now.

NRAM is non-volatile, like Flash memory, so your computer can boot up within a second. The chips would run at 2 GHz, and last virtually forever. The concept behind NRAM is simple: combine carbon nanotubes—a molecule of carbon atoms connected like a tube and sealed at both ends (see figure below)—with traditional semiconductor techniques. Ones and zeroes revisited: A nanotube is bent using electrostatic forces so that the top touches the bottom. Bent nanotubes can be unbent, too. Bent means a 1, and unbent means a 0. And billions of nanotubes can be squeezed into a four-inch wafer.

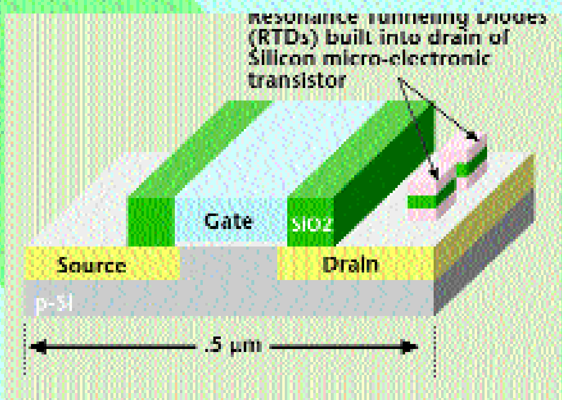
Nanotubes will find their application in high-resolution flat displays too—it's been discovered that nanotubes can emit ring-shaped electron spurts under certain conditions, and electron emission is basically what's required for a display device.



The Nanofuture Now

Alan Seabaugh, and his colleagues, of the Nanoelectronics Laboratory at Santa Clara University, USA, came up with an idea that could thrust us into the nanofuture right now—a hybrid transistor that belongs both in the micro and nano worlds. It's the Resonant Tunnelling Transistor (RTT) (*see figure 'Resonant Tunnelling Transistor'*). The RTDs increase the number of logic states that the transistor can be in, and this increases the 'logic density' on a chip. The RTT may be employed in commercial terabyte DRAM chips as early as 2007. Terabytes, not gigabytes.

With the shrinkage made possible by the RTT and other resonant tunnelling devices, we can expect Moore's law to continue unhindered for another generation!



We'll Live to See It Happen

If the last century was the century of the computer, this one could well be the century of nanotechnology. Some people say nanotechnology is 'the' technology of the future.

Advances are happening so rapidly, one can't tell when a major breakthrough is just round the corner. Keep an eye out for developments at sites such as ***kurzweilai.com***, ***eetimes.com*** and ***nanotech.org***.

Quantum computing is something else altogether: when it becomes a reality, it will change the way we look at reality forever. Closer to today, nanoelectronics is already promising to extend Moore's Law. Of course, there are nano-naysayers, but research dollars are pouring in by the barrel.

Nanotech promises a lot of things, but above all, it promises to make faster computing machines—a promise we'll probably live to see fulfilled. d

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